Mohamed Khalil **KHLIFI**

Mechatronics Engineer | Hardware Design & Electronics

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SUMMARY

Mechatronics Engineer specializing in hardware design and electronics, with hands-on experience in electronic circuit design, PCB layout, power electronics, motor control interfaces, and sensor integration.

PROFESSIONAL EXPERIENCE

Final Year Engineering Intern — Neuralbey — Monastir, Tunisia | March 2025 – September 2025 | View Source Code

- Architected an ESP32-based IoT smart pet feeder featuring automated dispensing, remote monitoring, scheduled feeding, a local UI and safety mechanisms.
- Integrated four sensors, an RTC module, stepper motor control, Blynk-based mobile communication and associated hardware interfaces.
- Laid out a custom PCB with power management circuitry and hardware interfaces for reliable system integration.

Key Technologies: ESP32 – Embedded C – IoT – Arduino IDE – PCB Design – Altium Designer – Wokwi – SolidWorks – Blynk – Sensors

Summer Engineering Intern — TAC–TIC — Ariana, Tunisia | July 2024 – September 2024 | View Source Code

- Developed an AI-powered YOLOv8 object-counting application for image, video, and webcam, with a Tkinter-based graphical UI for mode selection.

Key Technologies: Python – PyTorch – OpenCV – YOLOv8 – NumPy – PyCharm – Ultralytics – Tkinter – Computer Vision

Summer Engineering Intern — Relays and Industrial Automation (RAI) — Ariana, Tunisia | July 2023 – August 2023

- Analyzed industrial control schematics and supported component selection, electrical assembly, and PCB integration according to system requirements.
- Performed functional testing and troubleshooting to validate hardware operation and ensure system reliability.

Key Technologies: Industrial Automation – Electrical Schematics – Assembly – Functional Testing – PCB Assembly – Troubleshooting

PROJECTS

Stewart Platform (6-DOF Parallel Robot) | March 2026 | View Source Code

- Developed ESP32-based control for a 6-DOF Stewart Platform, coordinating six synchronized linear actuators with a 12 V / 10 A power architecture.
- Integrated ESP32, MPU6050 IMU, actuator interfaces and power circuitry into a unified control platform, validating connectivity and actuator synchronization.

Key Technologies: ESP32 – MPU6050 – Linear Actuators – FIT0805 – Hardware Integration – Electrical Architecture – Parallel Robotics – MATLAB

PCB & Circuit Design Projects | March 2025 – April 2025 | View Source Code

- Arduino Nano board: Recreated the board from schematics & datasheets, including PCB layout, component integration and electrical continuity verification.
- DC-DC converters (LM2577, LM2596, XL4015): Implemented buck and boost power-conversion circuits, from schematic to PCB layout and power-stage configuration, for reliable voltage regulation.
- Motor drivers (L293D, L298N, TB6612FNG): Developed H-Bridge motor drivers for embedded applications, applying PCB design principles and datasheet specifications for reliable DC motor control.
- IR sensor interfaces (FC-51, HW-006, KY-033, QTR-8RC): Designed sensor interface circuits for reliable signal acquisition and microcontroller compatibility.

Key Technologies: Altium Designer – PCB Layout – Schematic Capture – Power Electronics – Buck Converter – Boost Converter – H-Bridge – ATmega328P

Fighter Robot | March 2023 | View Source Code

- Built a combat robot controlled by an STM32, integrating a 36 V / 4.5 A electrical architecture for high-current BLDC motor operation and power distribution.
- Adapted hoverboard motherboard, motor controller, and RC receiver to interface with the robot's custom electronics and transmit control signals.

Key Technologies: Power Electronics – STM32F103RCT6 – BLDC Motor Control – Hardware Integration – RC Communication – Embedded C

All Terrain Robot & Line-Following Robot | February 2022 – May 2022 | View Source Code

- Assembled an all-terrain robot using the ESP32, BTS7960 for high-power DC motors control and PS3 Bluetooth wireless operation with stable power delivery.
- Fabricated an autonomous robot with five IR sensors, motor drivers, and Arduino control electronics for reliable line detection and stable navigation.

Key Technologies: ESP32 – BTS7960 – Bluetooth – PS3 Controller – Motor Control – PWM – Arduino – IR Sensors

EDUCATION

National Engineering School of Carthage (ENICarthage) | September 2021 – October 2025

- National Engineering Degree in Mechatronics Engineering (Master's equivalent)

Preparatory Institute for Engineering Studies of Sfax (IPEIS) | September 2019 – June 2021

- Engineering Preparatory Cycle – Physics and Technology

TECHNICAL SKILLS

Hardware & Electronics Design: Electronic Circuit Design – PCB Design – PCB Layout – Electrical Schematics – Board-Level Design – Component Selection – Electrical Architecture – Power Electronics – Footprint Creation – PCB Library Management – Datasheet Analysis – DRC/ERC – Electrical Assembly – Power Distribution – Schematic Capture

Embedded Hardware: STM32 – ESP32 – Arduino – Raspberry Pi – Hardware Bring-Up – Sensor Integration – Motor Driver Integration – Microcontroller Interfacing

Communication Interfaces: UART – I²C – SPI – CAN – Bluetooth – Wi-Fi – USB – Ethernet

Hardware Debugging & Validation: Oscilloscope – Digital Multimeter – Circuit Debugging – Hardware Testing & Troubleshooting – Through-Hole Soldering

Design Tools: Altium Designer – Proteus – SolidWorks

Programming Languages: Embedded C – C – Python

Development Tools: STM32CubeIDE – STM32CubeMX – Arduino IDE – Git – GitHub – VS Code

LANGUAGES

Arabic: Native proficiency **| English:** Professional Working Proficiency **| French:** Intermediate proficiency

ASSOCIATIVE ACTIVITIES

IEEE ENICarthage Student Branch / Mechatronics ENICarthage Club – Member **| October 2021 – December 2024**

ENICarthage Robots – Head of Line Following Committee **| February 2022**

ENICarthage Internal Contest, Line-Following Competition – 1st Place Winner **| January 2022**